

First Terminal Examination - 2079

Class : XI

Subject: Physics (1011)

Full Marks: 75

Time: 3 hrs.

SET 'B'

Pass Marks: 30

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

Group A

Circle the correct one from given alternatives.

[11 × 1 = 11]

- The force F is given by Stoke's law, $F = 6\pi\eta rv$ where r , v and η be the radius, velocity and coefficient of viscosity respectively. The correct dimensional formula of coefficient of viscosity is
 - $[ML^{-1}T^{-2}]$
 - $[ML^{-2}T^{-2}]$
 - $[ML^{-1}T^{-1}]$
 - $[MLT^{-1}]$
- The number of significant figures in the measurement 0.003 m are
 - 4
 - 3
 - 2
 - 1
- Which of the following quantity is dimensionless?
 - Stress
 - refractive index
 - angular velocity
 - universal gas constant
- A micrometer screw gauge of least count 0.01mm is used to measure the diameter of steel ball. The diameter is found to be 1cm. How should this measurement be recorded in the table of results?
 - 1.0 cm
 - 1.00cm
 - 1 cm
 - 1.000cm

5. The damping force 'F' on object moving with velocity 'v' in a medium is given by $F = Kv^2$, then the dimensional formula of K is
 - a. $[MT^{-2}]$
 - b. $[ML^{-1}]$
 - c. $[ML^2T^{-3}]$
 - d. $[MT^{-1}]$
6. At what temperature, reading on Celsius scale will be half the reading on Fahrenheit scale?
 - a. 120°C
 - b. 160°C
 - c. 160°F
 - d. 180°C
7. A light ray incident normally on a plane mirror suffers a deviation of
 - a. zero
 - b. $\frac{\pi^c}{2}$
 - c. π^c
 - d. $2\pi^c$
8. If a spherical mirror is immersed in liquid, its focal length
 - a. increases
 - b. decreases
 - c. remains constant
 - d. depends on nature of light
9. Two mirrors are inclined at 60° with each other. The number of images formed of an object is
 - a. 3
 - b. 4
 - c. 5
 - d. 6
10. The field of view is maximum for
 - a. plane mirror
 - b. concave mirror
 - c. convex mirror
 - d. cylindrical mirror
11. A spherical mirror produces an image of magnification '3'. Then the distance of the object from the mirror may be, if the focal length of spherical mirror is 24cm
 - a. 32cm only
 - b. 16cm only
 - c. 32cm, 24cm
 - d. 32cm, 16cm

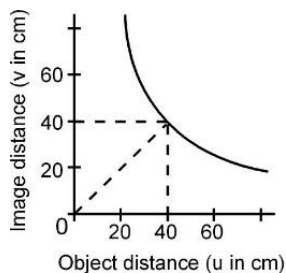
Group – B

Give short answer to the following questions.

[8 × 5 = 40]

12. a. In the wave equation $y = A \sin(\omega t - kx)$, where 't' is time and 'x' is distance, obtain the dimensional formula of ω and k. [2]
- b. Check the dimensional correctness of the equation $Fs = \frac{1}{2}m(v^2 - u^2)$, where 'F' is force, 's' is displacement, 'u' and 'v' are initial and final velocities. [2]
- c. Write two applications of dimensional analysis. [1]
13. a. What are significant figures? Find the number of significant figures in 0.020800 m and 103.0 kg. [2]
- b. The density of gold is 19.3 g/cm^3 . Express it into kg/m^3 using dimensional analysis. [2]
- c. Obtain the dimensional formula for universal gravitational constant. [1]
14. a. Define dimensions of a physical quantity? Write any two limitations of dimensional analysis. [1+2]
- b. Find the dimension of Planck's constant 'h' from the relation $E = hf$, where 'E' is energy and 'f' is frequency of photon. Also write down its unit. [2]
15. a. If you have a bucket of water and a cup of tea at the same temperature, which one of them has greater heat (assume both have nearly same specific heat capacity)? Explain. [2]
- b. Two beakers of water, A and B, initially are at the same temperature. The temperature of the water in beaker A is increased by 10°F and the temperature of the water in beaker B is increased by 10K. After these temperature changes,
- i) which beaker of water has the higher temperature? Explain. [2]
- ii) what about their heat energy? [1]

16. a. Two metallic rods of the same material but of different length are heated. Smaller rod has circular area of cross-section but larger rod has rectangular cross-section. [2+1]
- Will their linear expansivity be the same or different? Give justification of your answer.
 - What about the linear expansivity of same material having same dimension? Explain.
- b. A tumbler is the most generic of the serving glasses for liquids. Why does a thick glass tumbler sometimes crack when you pour very hot tea into it? [2]
17. A small object is placed at right angles to the axis of a fixed concave mirror so as to form (i) a real (ii) a virtual image, twice as long as the object. If the radius of the mirror is 40cm,
- Calculate the image distance in both cases. [2+2]
 - What is the distance between the two images? [1]
18. a. Write down one use of a convex mirror. [1]
- Write the differences between real and virtual images. [1]
 - An erect image, three times the size of the object is obtained with a concave mirror of radius of curvature 36 cm. What is the position of the object? [3]
19. a. Graph shows the variation of object distance and image distance of a concave mirror. Study the graph and answer the questions below:



- Calculate the focal length of concave mirror.

- ii. Calculate the magnification.
 - iii. How does the magnification of an object vary in concave mirror when object is moved from centre of curvature to focus? [3]
- b. Sketch the ray diagram to show the nature of image formed when object is placed at focus of concave mirror. [2]

Group – C

Give long answer to the following questions.

[3 × 8 = 24]

20. a. Using the method of dimension, show that the period of oscillation of simple pendulum is given by $T = 2\pi \sqrt{\frac{l}{g}}$, where symbols have their usual meanings. [3]
- b. Can the dimensionally correct equation be necessarily correct physically? What about dimensionally wrong equation? Explain this with suitable example. [2]
- c. The velocity 'v' of a particle at time 't' is given as $v = at + \frac{b}{(t + c)}$, where a, b, c, are constants. Find the dimensions of a, b & c. [3]
21. a. A thermometer is laid out in direct sunlight. Does it measure the temperature of air or of the sun or something else? Explain. [2]
- b. What do you mean by thermometric substance? Why mercury is widely used as thermometric substance in a liquid thermometer? [3]
- c. The temperature of the melting point of ice and that of boiling point of water at normal atmospheric pressure are marked 40 and 200 respectively on a certain thermometer. What is [3]
- i. the temperature in $^{\circ}\text{C}$ when the reading on this thermometer is 80.
 - ii. the thermometer reading when the temperature is 60°C .

22. a. What do you mean by principal focus of a convex mirror? [1]
- b. Prove mirror formula for a convex mirror. Can a convex mirror form a magnified image of real object? [3+1]
- c. A pencil is placed 6 cm in front of a concave mirror having a radius of curvature of 40 cm.
- i. Determine the position of the image formed by this mirror. [2]
- ii. Calculate magnification of the object. [1]

❧ - The End - ❧